

JOÃO MUÑOZ

SUMMARY

- +351 963 342132
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- Lisbon, Portugal
- [LinkedIn](#)
- [Github](#)

Software developer trained at **42** with strong focus on systems programming, **low-level** development, and **Unix environments**. Experienced in building complex projects in **C**, including shell implementation. Background in various **leadership roles** and **high-responsibility** field operations, **strong problem-solving skills**, **consistency**, autonomy, and **resilience**. Seeking junior backend, systems programming, or infrastructure-oriented software engineering opportunities.

WORK EXPERIENCE

Infraestruturas de Portugal (IP) Team Leader of Safety and Signalization - Linha AV Lisboa-Madrid (Lote C)

- Coordinated daily field operations ensuring compliance with safety protocols and project timelines
 - Managed risk assessment and rapid decision-making in **high-pressure** environments.
 - Recognized for **high productivity** and leadership within management team.
- Elvas, Portugal
2020 - 2023

Castilho Hostel and Suites 63 Receptionist - Developed Automation Pipeline

- Managed guest check-ins and issue resolution independently in a high-volume hostel environment.
 - Developed **strong communication** and problem-solving skills in dynamic environments.
 - **Developed a Python automation pipeline** to streamline operational workflows including reservations processing, guest communication, and access code management.
 - **Integrated Selenium, Excel-based workflows with openpyxl, WhatsApp Web**, and external services to **automate** repetitive manual tasks and **improve operational efficiency**. **Currently being used internally**.
 - Implemented **robust error handling** and automation reliability improvements for **production-like daily usage**.
- Lisbon, Portugal
2023 - 2025

EDUCATION

42 LISBOA Software Development

Part of the prestigious **42 network**, ranked among the top institutions for innovative education in the WURI (World's Universities with Real Impact) ranking.

Lisbon, Portugal
April 2025 - PRESENT

PROJECTS

Minishell — Unix Shell Implementation (C, Linux)

- Developed a Unix-like shell in **C** implementing process **creation**, **pipes**, **redirections**, and **signal handling**.
- Managed memory allocation and process control using low-level system calls (**fork**, **execve**, **wait**).

Nov 2025

Philosophers — Threads and Processes

- Implemented the Dining Philosophers problem in **C** using **POSIX threads**, focusing on synchronization and shared resources.
- Ensured **thread safety** with **mutexes**, preventing race conditions and deadlocks.

Oct 2025

CPP Modules (CPP0 – CPP9) — C++

- Developed **object-oriented** applications in **C++**, working with **inheritance**, **polymorphism**, memory management, **templates**, **STL containers**, and exception handling.
- Applied software engineering principles including abstraction, modular design, generic programming, and performance-conscious development.

Jan - April 2025

SKILLS

- **Programming:** C / C++ / Python
- **Systems:** Linux, Git
- **Concepts:** Memory management, Process control, Parsing, Raycasting

LANGUAGES

- **English (C1)**
- Portuguese (Native)
- Spanish (B2)